

Udit Karthikeyan



Email



Website



925-321-2217



Github



LinkedIn

Education

University of Illinois Urbana-Champaign

Aug 2022 - Dec 2024

Bachelor's of Science: Computer Science | GPA: 3.93

Coursework: Artificial Intelligence, Deep Learning, Real-Time Systems, Communication Networking

Experience

Amazon Web Services (AWS)

Feb 2025 - Current

SDE II: Amazon Data Firehose, EC2, Lambda, Akka, Kinesis, MCP, Bedrock

- Built a frontend load balancer system to improve system resilience under high load and failure scenarios, significantly reducing outage risk while delivering ~\$350K in annual cost savings
- Spearheaded team wide AI adoption efforts, creating a Firehose agent and orchestration tool and MCP server with Bedrock Knowledgebase used internally and by more than 5000 customers.
- Independently automated and launched critical service infrastructure in new regions, achieving a record-setting rollout (3x faster than historical timelines)

Coyoteland

May 2023 - Jan 2024

Full Stack Developer: Javascript (Puppeteer, React, Express, Node), Python, Zapier

- Built a Puppeteer-based land parcel scraper extracting more than 1 million data points for analytics
- Engineered a scalable Node.js/Express backend and React-based frontend for data insights
- Led a P2P marketing campaign contributing to ~200% sales growth

Freelance Website Designer

May 2022 - Jan 2025

Frontend Engineer: React, ThreeJS, CSS, Tailwind, Astro, Next.js

- Utilized React, Three.js, and complex JavaScript to enable interactive 3D modeling and real-time state manipulation across websites, including an AI travel agent and a web-based robot simulation tool
- Developed a phonetic learning web application for educators, featuring custom content management and interactive components to facilitate sight word pronunciation.
- Designed and delivered responsive personal websites for independent clients, managing the full development lifecycle from requirements gathering to deployment and hosting.

Projects

Bluefin

Built a Python-based desktop application that integrates with a real-estate data API to evaluate property deals, computing profitability, tax impact (REET), and pricing recommendations from user-defined queries

Mentra Chess

Developed an AR chess application for smart glasses using an open-source SDK, enabling hands-free gameplay, puzzle solving, and real-time matches against both Stockfish and other AR users. Designed low-latency input and rendering flows optimized for constrained AR hardware

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, OCaml/Haskell, Rust

Frameworks + Tools: React, GraphQL, Kubernetes, Akka, Node.js, Flask, LeRobot, PyTorch, Linux, ROS/ROS2, AWS, Jupyter, Postman, Bedrock, OpenCV